3. Prevention or Delay of Diabetes and Associated Comorbidities: Standards of Care in Diabetes—2026

*American Diabetes Association Professional Practice Committee for Diabetes**

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The American Diabetes Association (ADA) "Standards of Care in Diabetes" includes the ADA's current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA's clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

For guidelines related to screening for increased risk for type 1 diabetes, prediabetes and type 2 diabetes, and other forms of diabetes, please refer to section 2, "Diagnosis and Classification of Diabetes." For guidelines related to screening, diagnosis, and management of diabetes in children and adolescents, please refer to section 14, "Children and Adolescents."

Recommendations

3.1 In people with prediabetes, monitor for the development of diabetes at least annually; modify frequency of testing based on individual risk assessment. **E**

3.2 In people with presymptomatic type 1 diabetes, monitor for disease progression using A1C approximately every 6 months and 75-g oral glucose tolerance test (i.e., fasting and 2-h plasma glucose) annually; modify frequency of monitoring and consider augmenting with other glycemic assessment tools such as continuous glucose monitoring metrics based on individual risk assessment incorporating age, number and type of autoantibodies, and glycemic metrics. **E**

Screening for prediabetes (which refers to hyperglycemia preceding the diagnosis of type 2 diabetes only) and type 2 diabetes risk through an assessment of risk factors (**Table 2.5**) or with an assessment tool, such as the American Diabetes Association risk test, which can be used by either a layperson or a health care professional (diabetes.org/diabetes-risk-test), is recommended to guide whether to perform a diagnostic test for prediabetes (**Table 2.2**) and type 2 diabetes (**Table 2.1**) (see section 2, "Diagnosis and Classification of Diabetes"). Testing high-risk individuals for prediabetes is warranted because the laboratory assessment is safe and reasonable

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in cost, and early detection of hyperglycemia affords substantial time to intervene and delay or prevent the onset of type 2 diabetes and/or its long-term complications. Indeed, once identified, several effective therapeutic approaches exist that can delay type 2 diabetes in those with prediabetes with an A1C 5.7–6.4% (39–47 mmol/mol), impaired glucose tolerance (IGT) on 75-g oral glucose tolerance test (OGTT), or impaired fasting glucose (IFG). The utility of screening with A1C for prediabetes and diabetes may be limited in the presence of certain hemoglobinopathies and conditions that affect red blood cell turnover (**Table 2.3**). See section 2, "Diagnosis and Classification of Diabetes," and section 6, "Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises," for additional details on the appropriate use and limitations of A1C testing.

Three stages of type 1 diabetes have been defined, with symptomatic type 1 diabetes classified as stage 3 (**Table 2.4**). In individuals at risk for developing clinical type 1 diabetes, younger age of seroconversion (particularly under age 3 years), the total number of diabetes-related autoantibodies (1), and the development of autoantibodies against islet antigen 2 (IA-2) have all been associated with a more rapid progression to stage 3 type 1 diabetes. While continuous glucose monitoring can predict progression to stage 3 type 1 diabetes in children and adolescents with autoantibodies (2), OGTT-based metrics are more predictive of progression compared with continuous glucose monitoring (3). The decision to perform an OGTT may depend on such factors as eligibility and interest for stage-specific treatments, interest in participation in clinical research, OGTT availability, and the burden of testing. Consensus guidance provides expert recommendations on what should be monitored and how often in people with presymptomatic type 1 diabetes (4). Cost-effectiveness of population-based screening programs for type 1 diabetes has not been established (5,6), but such screening is being implemented outside the U.S. (7).

LIFESTYLE BEHAVIOR CHANGE FOR TYPE 2 DIABETES PREVENTION

Recommendations

3.3 Refer adults with overweight or obesity at high risk of type 2 diabetes to a diabetes prevention program to achieve and maintain a weight reduction of at least 5–7% of initial body weight through a healthy reduced-calorie eating pattern and ≥150 min/week of moderate-intensity physical activity. **A**

3.4 Prescribe an evidence-based eating pattern (e.g., Mediterranean, low carbohydrate) to individuals with prediabetes to prevent type 2 diabetes. **B**

3.5 Offer diabetes prevention programs to adults at high risk for type 2 diabetes. **A** Diabetes prevention programs should be covered by third-party payors, and inconsistencies in access should be addressed. **E**

3.6 Based on individual preference, certified technology-assisted diabetes prevention programs through smartphones, web-based applications, and telehealth can be effective in preventing type 2 diabetes and should be considered. **B**

The Diabetes Prevention Program

Several major randomized controlled trials, including the Diabetes Prevention Program (DPP) (8), the Finnish Diabetes Prevention Study (DPS) (9), and the Da Qing Diabetes Prevention Study (Da Qing study) (10), demonstrated that lifestyle/behavioral intervention with an individualized reduced-calorie meal plan is highly effective in preventing or delaying type 2 diabetes and improving other cardiometabolic risk factors such as blood pressure, lipids, and markers of inflammation (11). The strongest evidence for diabetes prevention in the U.S. comes from the DPP trial (8). The DPP demonstrated that intensive lifestyle intervention could reduce the risk of incident type 2 diabetes by 58% over 3 years. Follow-up of three large trials of lifestyle intervention for diabetes prevention showed sustained reduction in the risk of progression to type 2 diabetes: 39% reduction at 30 years in the Da Qing study (12), 43% reduction at 7 years in the Finnish DPS (9), and 34% reduction at 10 years (13), 27% reduction at 15 years (14), and 24% reduction at 21 years in the U.S. Diabetes Prevention Program Outcomes Study (DPPOS) (15,16).

The DPP lifestyle intervention was a goal-based intervention. A complete description of the DPP and how to access it is available online at coveragetoolkit.org/about-national-dpp/ndpp-overview/. Briefly, the DPP administered a 16-session structured core curriculum, completed within the first 6 months of the program, followed by an additional 6 months of a flexible maintenance program. The core curriculum included sessions on reducing calories, increasing physical activity, self-monitoring, maintaining healthy lifestyle behaviors (such as how to make healthy choices when eating out), and guidance on managing psychological, social, and motivational challenges (17).

Two major goals of the DPP were to achieve and maintain at least 7% weight loss and to obtain 150 min of moderate-intensity physical activity per week, such as brisk walking. Weight loss was the most important factor in reducing incident diabetes. Obtaining at least 150 min of physical activity per week, even without achieving the weight loss goal, reduced incidence of type 2 diabetes by 44% (18). Participants were encouraged to distribute physical activity throughout the week with a minimum frequency of three times per week and at least 10 min per session. A maximum of 75 min of strength training could be applied toward the total 150 min/week physical activity goal. The initial focus of the nutrition intervention was on reducing total fat intake rather than caloric restriction. However, the concepts of calorie balance and the need to restrict total calories and fat were also introduced. Overall, the DPP's lifestyle intervention was designed to produce a 700 kcal/week energy deficit (17).

The 7% weight loss goal was selected because it was feasible to achieve and maintain, hypothesized to reduce incident diabetes risk, and improved other cardiometabolic risk factors. Further analysis suggested greater benefit for diabetes prevention with at least 7–10% weight loss with lifestyle interventions (18). The recommended pace of weight loss was 1–2 lb/week.

While the DPP interventions were successful in preventing or delaying the onset of type 2 diabetes, major cardiovascular events were not significantly different 21 years after the DPP concluded, which may be due to the widespread use of statin, antihypertensive, and metformin therapy among all participants well after the conclusion of the study (19). However, the incidence of diabetes continued to be significantly

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lower at 21 years of follow-up in both intensive lifestyle therapy and metformin groups, when compared with placebo, corresponding to increases in median diabetes-free survival of 3.5 years and 2.5 years, respectively (15). Importantly, the overall treatment effect was driven almost entirely by benefit seen during the initial DPP phase (15), underscoring the importance of early intervention and continued implementation of diabetes prevention efforts. Still, individuals can derive benefit from lifestyle interventions to prevent progression of prediabetes to diabetes irrespective of the duration of prediabetes prior to initiation of lifestyle intervention (20). Thus, there is clear potential benefit in diabetes prevention and minimal to no risk of harm with these interventions.

Nutrition

Nutrition counseling for weight loss in the DPP lifestyle intervention arm focused on reduction of total fat and calories (8,17,18). However, current evidence suggests there is no ideal percentage of calories from carbohydrate, protein, and fat to prevent diabetes; therefore, macronutrient distribution should be based on an individualized assessment of current eating patterns, preferences, and metabolic goals. A variety of eating patterns (21) can be appropriate for individuals with prediabetes or diabetes, including Mediterranean-style, plant-based (which may or may not be omnivorous), Dietary Approaches to Stop Hypertension (DASH), and low-carbohydrate (22–25). The overall quality of food consumed (as measured by the Healthy Eating Index, Alternative Healthy Eating Index, and DASH score), with an emphasis on whole grains, legumes, nuts, fruits, and vegetables and minimal refined and processed foods, is also associated with a lower risk of type 2 diabetes (26–28). For people at risk for or living with diabetes, individualized medical nutrition therapy (see section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes," for more detailed information) is effective in lowering A1C (29,30).

Physical Activity

Moderate-intensity physical activity, such as brisk walking for 150 min/week, has shown beneficial effects in those with prediabetes (8). Similarly, moderate-intensity physical activity has been shown to improve insulin sensitivity and reduce abdominal fat in children, adolescents, and young adults (31,32). Health care professionals are encouraged to promote a DPP-style program to all individuals who have been identified to be at an increased risk of type 2 diabetes. In addition to aerobic activity, a physical activity plan designed to prevent diabetes can include resistance training (17,33). Breaking up prolonged sedentary time may also be encouraged, as it lowers postprandial glucose levels (34). The effects of regular physical activity participation appear to extend to the prevention of gestational diabetes mellitus (GDM) as well (35,36).

Delivery and Dissemination of Lifestyle Behavior Change for Diabetes Prevention

Because the intensive lifestyle intervention in the DPP was effective in preventing type 2 diabetes among those at high risk, and both DPP and adapted lifestyle behavior change programs for diabetes prevention were shown to be cost-effective across multiple applications, broader efforts to disseminate scalable DPP-like lifestyle behavior change programs for diabetes prevention with coverage by third-party payors ensued (37–42). Group delivery of DPP content in community or primary care settings has demonstrated the potential to reduce overall program costs while still producing weight loss and diabetes risk reduction (43,44). Interventions to prevent type 2 diabetes can also be delivered through workplace programming, with interventions that most closely adhere to DPP content shown to be most effective (45).

The Centers for Disease Control and Prevention (CDC) developed the National Diabetes Prevention Program (National DPP), a resource designed to bring such evidence-based lifestyle change programs for preventing type 2 diabetes to communities (cdc.gov/diabetes-prevention). To be eligible for this program, individuals must have a BMI in the overweight range and be at risk for diabetes based on laboratory testing, a previous diagnosis of GDM, or a positive risk test (cdc.gov/prediabetes/risktest/). The CDC has also developed the Diabetes Prevention Impact Tool Kit (nccd.cdc.gov/toolkit/diabetesimpact) to help organizations assess the economics of providing or covering the National DPP (46). To expand preventive services using a cost-effective model, the Centers for Medicare & Medicaid Services expanded Medicare reimbursement coverage for the National DPP to organizations recognized by the CDC that become Medicare suppliers for this service (innovation.cms.gov/innovation-models/medicare-diabetes-prevention-program). The locations of Medicare DPPs are available online at innovation.cms.gov/innovation-models/medicare-diabetes-prevention-program/mdpp-map. To qualify for Medicare coverage, individuals must have BMI >25 kg/m^2 (or BMI >23 kg/m^2 if self-identified as Asian) and glycemic testing consistent with prediabetes in the last year. Medicaid DPP coverage is also expanding on a state-by-state basis.

While CDC-recognized behavioral counseling programs, including Medicare DPP services, have met minimum quality standards and are reimbursed by many payors, lower retention rates have been reported for younger adults and racial and ethnic minoritized populations (47). Therefore, other programs and modalities of behavioral counseling for diabetes prevention may also be appropriate and efficacious based on individual preferences and availability. Use of community health workers to support DPP-like interventions can be appropriate and cost-effective (48,49) (see section 1, "Improving Care and Promoting Health in Populations," for more information). The use of community health workers may facilitate the adoption of behavior changes for diabetes prevention while bridging barriers related to social determinants of health. However, coverage by third-party payors remains limited. Counseling by a registered dietitian nutritionist has been shown to help individuals with prediabetes improve eating habits, increase physical activity, and achieve 7–10% weight loss (50–53). Individualized medical nutrition therapy (see section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes," for more detailed information) is also effective in improving glycemia in individuals diagnosed with prediabetes (51,54). Furthermore, trials involving medical nutrition therapy for adults with prediabetes found significant reductions in weight, waist circumference, and glycemia (54,55). Individuals with prediabetes can benefit

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from referral to a registered dietitian nutritionist for individualized medical nutrition therapy upon diagnosis and at regular intervals throughout their treatment plan (52,56). Other health care professionals, such as pharmacists and diabetes care and education specialists, can also be considered for diabetes prevention efforts (56,57).

Technology-assisted programs may effectively deliver a DPP-like intervention (58–61). Such technology-assisted programs may deliver content through smartphones, web-based applications, and telehealth and may be an acceptable and efficacious option to bridge barriers, particularly for individuals with low income and/or in rural locations; however, not all technology-assisted programs are effective (58,62–64). The CDC Diabetes Prevention Recognition Program (DPRP) (https://www.cdc.gov/diabetes-prevention/php/program-provider/program-requirements.html) certifies technology-assisted modalities as effective vehicles for DPP-based interventions; such programs must use an approved curriculum, include interaction with a coach, and attain the DPP outcomes of participation, physical activity reporting, and weight loss. Health care professionals should consider referring adults with prediabetes to certified technology-assisted programs.

The CDC's Division of Diabetes Translation National DPP toolkit (coveragetoolkit.org/recruitment-referral-for-the-national-dpp-lifestyle-change-program) provides diabetes care team members resources for how to increase referral and uptake of the DPP. Some ways to increase DPP utilization include direct mailings, phone scripts, and text-based outreach. Using multiple methods of outreach to increase awareness among those at risk, and among health care professionals, in addition offering the DPP virtually can improve participation and uptake (65,66). However, more work is needed to improve uptake among high-risk populations (67,68).

Sleep Characteristics Associated With Increased Risk of Type 2 Diabetes

Sleep occupies approximately one-third of the day for most people and modulates important metabolic, endocrine, and cardiovascular processes (69). Sleep can be characterized using three key constructs: quantity, quality, and timing (i.e., chronotype). There is now established evidence for a U-shaped association between sleep duration and type 2 diabetes incidence, with the nadir typically occurring at 7 h per day, with short (typically defined as <6 h) and long (typically defined as >9 h) sleep duration having up to a 50% increase in the risk of type 2 diabetes, including progression from prediabetes (70). Sleep quality is defined as "an individual's self-satisfaction with all aspects of the sleep experience" (71,72). A meta-analysis reported that poor overall sleep quality was associated with a 40% increased risk of developing type 2 diabetes (73). Chronotype preference has been linked with many chronic diseases, including type 2 diabetes. For example, for those with a preference for evenings (i.e., going to bed late and getting up late), there was a 2.5-fold higher odds ratio for type 2 diabetes than for those with a preference for mornings (i.e., going to bed early and getting up early), independent of sleep duration and sleep sufficiency (74). More detailed information concerning sleep and the impact of sleep on prediabetes and diabetes risk and management can be found in section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes."

PHARMACOLOGIC INTERVENTIONS TO DELAY TYPE 2 DIABETES

Recommendations

3.7 Metformin for the prevention of type 2 diabetes should be considered in adults at high risk of type 2 diabetes, as typified by the Diabetes Prevention Program, especially those aged 25–59 years with BMI ≥35 kg/m^2, higher fasting plasma glucose (e.g., ≥110 mg/dL [≥6 mmol/L]), and higher A1C (e.g., ≥6.0% [≥42 mmol/mol]), and in individuals with prior gestational diabetes mellitus. **A**

3.8 Consider using metformin to prevent hyperglycemia in high-risk individuals treated with a phosphatidylinositol 3-kinase α (PI3Kα) inhibitor (e.g., alpelisib and inavolisib). **B**

3.9 Consider using metformin to prevent hyperglycemia in high-risk individuals treated with high-dose glucocorticoids. **B**

3.10 Consider periodic assessment of vitamin B12 levels in individuals receiving long-term metformin therapy, especially in those with anemia or peripheral neuropathy. **B**

Because weight loss and maintenance through lifestyle (e.g., nutrition and physical activity) behavior changes might not be sufficient on their own (13), some people at high risk for type 2 diabetes may benefit from additional pharmacotherapy support. Various pharmacologic agents used to treat diabetes have also been evaluated for diabetes prevention. Metformin, α-glucosidase inhibitors, incretin receptor agonists (e.g., liraglutide, semaglutide, and tirzepatide), thiazolidinediones, and insulin have all been shown to lower the incidence of diabetes in specific populations (75–83). However, whether initiating glucose-lowering treatment early—before hyperglycemia reaches the diagnostic threshold of diabetes—improves long-term health outcomes is unknown. Indeed, the benefits of glucose lowering to the normoglycemic range need to be balanced against the potential adverse effects of these therapies and the burden and cost of treatment. There are currently no long-term data to support the use of pharmacologic treatments other than metformin for the sole purpose of preventing type 2 diabetes. However, use of glucagon-like peptide 1–based therapies for weight management in individuals with overweight or obesity—an essential component of type 2 diabetes prevention—is highly beneficial and should be considered, as discussed in detail in section 8, "Obesity and Weight Management for the Prevention and Treatment of Diabetes." Importantly, in the DPP, weight loss was key to reducing the risk of progression to diabetes, with every kilogram of weight loss conferring a 16% reduction in risk of progression over 3.2 years (18). In individuals with previous history of GDM, the risk of type 2 diabetes increased by 18% for every 1-unit increase in BMI above the preconception baseline (84). Several medications evaluated for obesity treatment (e.g., orlistat, phentermine and topiramate, liraglutide, semaglutide, and tirzepatide) have been shown to decrease the incidence of type 2 diabetes in those with prediabetes (80,85–87).

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Other medications have also been assessed for their efficacy in preventing type 2 diabetes. The antihypertensive valsartan was found to modestly reduce the incidence of diabetes among individuals with prediabetes and cardiovascular disease or risk factors, but with no difference in cardiovascular outcomes (88). Thus, we do not recommend using valsartan for the prevention of diabetes. Testosterone therapy also reduced the risk of incident type 2 diabetes by 41% in men aged 50–74 years with central obesity and prediabetes (89). However, the benefit of testosterone was observed only when added to a structured lifestyle program. Indeed, the TRAVERSE Diabetes Study (a prespecified efficacy trial nested within the large multicenter TRAVERSE trial whose primary objective was to assess the cardiovascular outcomes of testosterone therapy in men aged 45–80 years with hypogonadism) found no benefit of testosterone as compared with placebo with respect to progression from prediabetes to diabetes or remission of diabetes among those who had prediabetes or diabetes at baseline, respectively (90). Testosterone therapy also did not alter glucose or A1C levels among participants with prediabetes or diabetes, nor were there any differences in outcomes by baseline testosterone level, preexisting cardiovascular disease, age, or race. Thus, we do not recommend testosterone therapy for the prevention of type 2 diabetes in men with hypogonadism; the decision to initiate testosterone replacement needs to be informed by individualized treatment considerations that weigh the expected benefits and harms of treatment.

There has been significant interest in understanding the potential of vitamin D therapy to prevent progression of high-risk prediabetes to type 2 diabetes in adults (91). Three randomized controlled trials tested whether vitamin D therapy in combination with lifestyle modification reduces the risk of developing diabetes in adults with high-risk prediabetes (i.e., IGT or meeting two or three criteria for the diagnosis of prediabetes [fasting glucose, A1C, 2-h glucose after a 75-g OGTT]): the Tromsø study in Norway, with 511 participants; the Vitamin D and Type 2 Diabetes (D2d) study in the U.S., with 2,423 participants; and the Diabetes Prevention with Active Vitamin D (DPVD) study in Japan, with 1,256 participants (92–94). Although vitamin D therapy consistently appeared to modestly reduce the risk of developing diabetes in all three trials, none of the results were statistically significant, which was attributed to insufficient statistical power. Subsequently, several meta-analyses pooling the three randomized clinical trials along with other smaller studies suggested a modest (i.e., 10–15% risk reduction) potential benefit in specific populations, such as older adults, individuals with lower baseline vitamin D levels, and individuals without obesity (95,96). However, there are several concerns and uncertainties regarding recommending widespread vitamin D therapy for adults with high-risk prediabetes. First, these trials used varying dosages of vitamin D that were higher than the recommended daily allowance (i.e., 600 IU/day for those aged 18–70 years and 800 IU/day for those older than 70 years). Second, the benefit-to-risk ratio of vitamin D therapy for people with high-risk prediabetes remains uncertain. Although there were no safety concerns identified in these trials, the anticipated treatment of many millions of adults with prediabetes in the U.S. and globally could increase the risk of adverse events (such as hypercalcemia, hypercalciuria, nephrolithiasis, and kidney failure), especially treatment with unspecified doses of vitamin D without monitoring blood 25-hydroxy vitamin D levels.

Thus, the expected benefit, harm, treatment burden, and cost of using any medication to prevent type 2 diabetes must be considered for each individual as part of informed shared decision-making. This is particularly important because any pharmacologic intervention is likely to be needed to be used long-term because of waning effects after stopping the medication. At the present time, there are no pharmacologic agents approved by the U.S. Food and Drug Administration specifically for the prevention of type 2 diabetes.

Metformin

Metformin has the most robust efficacy and safety data as a pharmacologic therapy for diabetes prevention among people with prediabetes (82). Importantly, metformin was overall less effective than lifestyle modification in the DPP, although group differences attenuated over time in the DPPOS (14) and metformin was as effective as lifestyle modification in participants with BMI $\geq$35 kg/m^2 and in younger participants aged 25–44 years (8). In individuals with a history of GDM, metformin and intensive lifestyle modification led to an equivalent 50% reduction in diabetes risk (97). Both interventions remained effective during a 10-year follow-up period (98), with lifestyle therapy being highly cost-effective ($12,878 per quality-adjusted life-year [QALY]) and metformin marginally cost-saving compared with placebo (37). By the time of the 15-year follow-up (DPPOS), exploratory analyses demonstrated that participants with a higher baseline fasting glucose ($\geq$110 mg/dL [$\geq$6 mmol/L] vs. 95–109 mg/dL [5.3–5.9 mmol/L]), those with a higher A1C (6.0–6.4% [42–46 mmol/mol] vs. <6.0% [<42 mmol/mol]), and individuals with a history of GDM (vs. individuals without a history of GDM) experienced higher risk reductions with metformin, identifying subgroups of participants that may benefit the most from metformin (83). In the Indian Diabetes Prevention Program (IDPP-1), metformin and lifestyle intervention reduced diabetes risk similarly at 30 months; however, the lifestyle intervention in IDPP-1 was less intensive than that in the DPP (99). Based on findings from the DPP, metformin should be recommended as an option for high-risk individuals (e.g., younger individuals, those with history of GDM, or those with BMI $\geq$35 kg/m^2).

Individuals receiving high-dose and/or long-duration glucocorticoid therapy are at risk for developing glucocorticoid-induced diabetes (see section 2, "Diagnosis and Classification of Diabetes"). Metformin can be considered to prevent the development of glucocorticoid-induced hyperglycemia in high-risk individuals, notably those receiving higher doses of glucocorticoids, with longer treatment duration, and in the presence of other risk factors for diabetes. The efficacy of metformin to prevent hyperglycemia—particularly insulin resistance and postprandial hyperglycemia—has been demonstrated in small randomized clinical trials both in healthy individuals (100) and in individuals receiving high-dose glucocorticoids for the treatment of cancer (101) and rheumatologic/autoimmune disorders (102,103). In one trial with 40 participants receiving high-

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dose prednisolone for the treatment of inflammatory conditions, metformin also reduced rates of infection and all-cause hospital readmissions (103).

Proactive use of metformin may also be beneficial in preventing hyperglycemia in individuals receiving phosphatidylinositol 3-kinase α (PI3Kα) inhibitors (e.g., alpelisib and inavolisib) for the treatment of cancer (see section 2, "Diagnosis and Classification of Diabetes," for additional details on diabetes induced by systemic anticancer therapy). This is particularly important because hyperglycemia is the most frequent adverse event of PI3Kα inhibitor therapy that leads to treatment discontinuation (104). In a phase 2 trial of metformin to prevent hyperglycemia in individuals receiving alpelisib for advanced breast cancer (metformin was started 1 week prior to starting alpelisib and continued for the duration of treatment), metformin prevented the development of grade 3–4 hyperglycemia in all but four participants, and none had to discontinue alpelisib therapy (105). Similar benefits were seen with implementation of a metformin hyperglycemia-prevention protocol in routine practice (106). Individuals at highest risk for developing hyperglycemia with PI3Kα inhibitor therapy include those who are older (≥70 years old), with obesity (BMI ≥30 kg/m^2), concurrently treated with glucocorticoids, and with hyperglycemia at baseline (e.g., A1C ≥5.7% [39 mmol/mol] or fasting plasma glucose ≥100 mg/dL [5.6 mmol/L]) (107, 108). However, it is important to weigh the benefits of preventing hyperglycemia with the potential adverse effects of metformin therapy, particularly diarrhea, which is also a frequent adverse effect of PI3Kα therapy (though there is no evidence that concurrent use of these two medications further exacerbates diarrhea risk) (108).

Decreased vitamin B12 levels are a known consequence of long-term treatment with metformin (109). Periodic assessment of vitamin B12 level in those taking metformin chronically should be considered to check for possible deficiency, especially in those receiving a higher dose (e.g., ≥1,500 mg/day) (110), with longer treatment duration (4–5 years) (111), with other risk factors for vitamin B12 deficiency, and if a deficiency is suspected, such as in people with anemia, peripheral neuropathy, or chronic kidney disease (109,112,113) (see section 9, "Pharmacologic Approaches to Glycemic Treatment," for more details). A person who has been taking metformin for more than 4 years or is at risk for vitamin B12 deficiency for other reasons (e.g., vegan dietary pattern, previous gastric/small bowel surgery) should be monitored for vitamin B12 deficiency annually (114,115).

PREVENTION OF VASCULAR DISEASE AND MORTALITY

Recommendations

3.11 Prediabetes is associated with heightened cardiovascular risk; therefore, screening for and treatment of modifiable risk factors for cardiovascular disease are suggested. **B**

3.12 Statin therapy may increase the risk of type 2 diabetes in people at high risk of developing type 2 diabetes. In such individuals, glucose status should be monitored regularly and diabetes prevention approaches reinforced. It is not recommended that statins be avoided or discontinued for this adverse effect. **B**

3.13 In people with a history of stroke and evidence of insulin resistance and prediabetes, pioglitazone may be considered to lower the risk of stroke or myocardial infarction. However, this benefit needs to be balanced with the increased risk of weight gain, edema, and fractures. **A** Lower doses may mitigate the risk of adverse effects but may be less effective. **C**

People with prediabetes often have other cardiovascular risk factors, including hypertension and dyslipidemia (116), and are at increased risk for cardiovascular disease (117,118). Evaluation for tobacco use and referral for tobacco cessation should be part of routine care for those at risk for diabetes. Of note, the years immediately following smoking cessation may represent a time of increased risk for diabetes (119,120), and individuals should be monitored for diabetes development and receive evidence-based lifestyle behavior change for diabetes prevention as described in this section. See section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes," for more detailed information. The lifestyle interventions for weight loss in study populations at risk for type 2 diabetes have shown a reduction in cardiovascular risk factors and the need for medications used to treat these cardiovascular risk factors (121,122). The lifestyle intervention in the Da Qing study was associated with lowering cardiovascular disease and mortality at 23 and 30 years of observational follow-up (10,12). Treatment goals and therapies for hypertension and dyslipidemia in the primary and secondary prevention of cardiovascular disease for people with prediabetes should be based on their level of cardiovascular risk. Increased vigilance is warranted to identify and treat these and other cardiovascular disease risk factors (123).

Statin use increases risk of diabetes (124,125). In the DPP, statin use was associated with greater diabetes risk irrespective of the treatment group (pooled hazard ratio [HR] [95% CI] for incident diabetes 1.36 [1.17–1.58]) (124). In trials of primary and secondary prevention of cardiovascular disease, cardiovascular and mortality benefits of statin therapy exceed the risk of diabetes (126,127), suggesting a highly favorable benefit-to-harm balance with statin therapy. Hence, discontinuation of statins due to concerns of diabetes risk is not recommended in this population.

Results of cardiovascular outcome trials in people without diabetes also provide important evidence for cardiovascular risk reduction in people with increased cardiometabolic risk, though not specifically prediabetes (see section 10, "Cardiovascular Disease and Risk Management," for more details). The IRIS (Insulin Resistance Intervention after Stroke) trial of people with a recent (<6 months) stroke or transient ischemic attack, without diabetes but with insulin resistance (as defined by a HOMA of insulin resistance index of ≥3.0), evaluated pioglitazone (goal dose of 45 mg daily) compared with placebo. At 4.8 years, the risk of stroke or myocardial infarction, as well as the risk of diabetes, was lower in the pioglitazone group than in the placebo group; weight gain, edema, and fractures were higher in the pioglitazone treatment group (128–130). Lower doses may mitigate the adverse effects but may also be less effective (131).

PERSON-CENTERED CARE GOALS

Recommendations

3.14 In adults with overweight or obesity at high risk of type 2 diabetes, care goals should include weight

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loss and maintenance, minimizing the progression of hyperglycemia, and attention to cardiovascular risk. **B**

3.15 Pharmacotherapy (e.g., for weight management, minimizing the progression of hyperglycemia, and cardiovascular risk reduction) should be considered to support person-centered care goals. **A**

3.16 More intensive preventive approaches should be considered in individuals who are at particularly high risk of progression to diabetes, including individuals with BMI ≥35 kg/m^2, those with higher glucose levels (e.g., fasting plasma glucose 110–125 mg/dL [6.1–6.9 mmol/L], 2-h postchallenge glucose 173–199 mg/dL [9.6–11.0 mmol/L], and A1C ≥6.0% [≥42 mmol/mol]), and individuals with a history of gestational diabetes mellitus. **A**

Individualized risk-to-benefit ratio should be considered in screening, intervention, and monitoring to lower the risk of type 2 diabetes and associated comorbidities. Multiple factors, including age, BMI, and other comorbidities, may influence the risk of progression to diabetes and lifetime risk of complications (132,133). Prediabetes is associated with increased cardiovascular disease and mortality (118), which emphasizes the importance of attending to cardiovascular risk in this population. However, the new diagnosis of prediabetes in older adults (aged >70 years) is less relevant for progression to diabetes; indeed, regression to normoglycemia or death was more frequent than progression to diabetes in the Atherosclerosis Risk in Communities (ARIC) study (132). Moreover, diabetes-related morbidity and mortality is lower and the benefits of treatment are not established in those who develop type 2 diabetes after age 70 years (134).

In the DPP, which enrolled high-risk individuals with IGT, elevated fasting glucose, and elevated BMI, the crude incidence of diabetes within the placebo group was 11 cases per 100 person-years, with a cumulative 3-year incidence of diabetes of 29% (8). Characteristics of individuals in the DPP and DPPOS who were at particularly high risk of progression to diabetes (crude incidence of diabetes 14–22 cases per 100 person-years) included BMI ≥35 kg/m^2, higher glucose levels (e.g., fasting plasma glucose 110–125 mg/dL [6.0–6.9 mmol/L], 2-h postlenge glucose 173–199 mg/dL [9.6–11.0 mmol/L], A1C ≥6.0% [≥42 mmol/mol]), or a history of GDM (8,97,98). In contrast, in the community-based ARIC study, observational follow-up of adults with mean age 75 years with laboratory evidence of prediabetes (based on A1C 5.7–6.4% [39–47 mmol/mol] and/or fasting glucose 100–125 mg/dL [5.6–6.9 mmol/L]), but not meeting specific BMI criteria, found lower progression to diabetes over 6 years: 9% of those with A1C-defined prediabetes and 8% of those with IFG (133).

Thus, it is important to individualize the risk-to-benefit ratio of intervention and consider person-centered goals. Risk models have generally found a higher benefit of the intervention in those at highest risk (18). Diabetes prevention trials and observational studies highlight key principles that may guide person-centered goals. In the DPP, which enrolled a high-risk population meeting criteria for overweight or obesity, weight loss was an important mediator of diabetes prevention or delay, with greater metabolic benefit seen with greater weight loss (18,135). In the DPP and DPPOS, progression to diabetes, duration of diabetes, and mean level of glycemia were important determinants of the development of microvascular complications (14). Achieving normal glucose regulation, even once, during the DPP was associated with a lower risk of diabetes and lower risk of microvascular complications irrespective of the treatment arm (136). Observational follow-up of the Da Qing study also showed that regression from IGT to normal glucose tolerance or remaining with IGT rather than progressing to type 2 diabetes at the end of the 6-year intervention trial resulted in significantly lower risk of cardiovascular disease and microvascular disease over 30 years (137).

Pharmacotherapy for weight management and cardiovascular risk reduction (see section 8, "Obesity and Weight Management for the Prevention and Treatment of Diabetes," and section 10, "Cardiovascular Disease and Risk Management," for more details) can be considered to support individualized person-centered goals, with more intensive preventive approaches considered in individuals at high risk of progression.

PREVENTION OR DELAY OF SYMPTOMATIC TYPE 1 DIABETES

Lifestyle and Type 1 Diabetes Progression

Observational studies have identified several factors that increase β-cell demand and risk of progression to clinical type 1 diabetes among individuals with islet autoantibodies, including lower levels of physical activity (138), higher glycemic index eating patterns (139), and total sugar intake (140). Similar associations have not been observed for the development of autoantibodies. In The Environmental Determinants of Diabetes in the Young (TEDDY) longitudinal study, daily minutes spent in moderate to vigorous physical activity were associated with a reduced risk of progression to type 1 diabetes in children 5–15 years of age with multiple islet autoantibodies (HR 0.92 [95% CI 0.86–0.99] per 10-min increase; P = 0.02) (138). In the Diabetes Autoimmunity Study in the Young (DAISY), also in children with islet autoantibodies, consumption of higher glycemic index foods (HR 2.20 [95% CI 1.17–4.15]) and total sugar intake (HR 1.75 [95% CI 1.07–2.85]) (139,140) were both associated with progression to type 1 diabetes. In nonobese diabetic mice, an animal model for the development of type 1 diabetes, sustained high-glucose drinking significantly aggravated islet inflammation and accelerated the onset of type 1 diabetes (141). However, efficacy of lifestyle interventions that modify these factors in individuals with stage 1 or stage 2 type 1 diabetes has not yet been reported.

Pharmacologic Interventions to Delay Symptomatic Type 1 Diabetes

Recommendation

3.17 Teplizumab-mzwv infusion to delay the onset of symptomatic type 1 diabetes (stage 3) should be discussed with selected individuals aged ≥8 years with stage 2 type 1 diabetes. Treatment should be in a setting with appropriately trained personnel. **B**

Teplizumab, a CD3-directed humanized monoclonal antibody engineered to have decreased Fc receptor binding, has been

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approved to delay the onset of stage 3 type 1 diabetes in people 8 years and older with stage 2 type 1 diabetes based in part on the results of a single trial in relatives of people with type 1 diabetes (142). In this study, 44 individuals were randomized to a 14-day course of teplizumab and 32 to placebo. The median time to stage 3 type 1 diabetes diagnosis was 48.4 months in the teplizumab group and 24.4 months in the placebo group. Type 1 diabetes was diagnosed in 19 (43%) participants who received teplizumab and 23 (72%) participants who received placebo (HR 0.41 [95% CI 0.22–0.78]). In prespecified analyses, the presence of HLA-DR4, absence of HLA-DR3, and absence of antizinc transporter 8 antibody predicted response to teplizumab (HR 0.20 [95% CI 09.0.45], 0.18 [0.07–0.45], and 0.07 [0.02–0.26], respectively). The most common adverse reactions were transient lymphopenia (73%) followed by rash (36%).

Numerous clinical studies are being conducted to test methods for preventing or delaying the onset of stage 3 type 1 diabetes in those with evidence of autoimmunity without symptoms or for delaying loss of insulin secretory capacity after onset of stage 3, some with promising results (see ClinicalTrials.gov and TrialNet.org).

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